

HelixCode Feature Status

HelixCode — Feature Status Summary

Revision: 1 · **Last modified:** 2026-07-01T19:45:00Z

Two-audience companion (§11.4.56) to [Status.md](#) for release codeserver-1.0.0-dev-0.0.3. Page 1 is for the project / product team; Page 2 is for software engineers. Always in sync with Status.md (§11.4.60).

Page 1 — For the project & product team

What HelixCode is. Browser-based VS Code that opens your real projects, behind a secure HTTPS front door, running without root — and now tied to your **real host account with your SSH key** (no password anywhere).

What works now (proven this release):

- **Log in with your SSH key, no password.** The login page shows a one-time challenge; you sign it locally with your key and paste it back. Wrong or missing signatures are refused. Proven end-to-end on the live system.
- **The editor is really you.** Once in, the editor runs as your host account, so your SSH keys, your .bashrc settings, and all your tools work in the terminal — including git over SSH. Proven live (a real git remote check returned results).
- **It stays locked if the guard goes down.** If the login guard stops, access is **denied**, never accidentally opened. Proven by killing the guard live and confirming access was blocked, then recovered.
- **Install and reboot survival.** One script installs everything without root; the services are set to come back after a reboot.

- **Plugins install from the marketplace.** A real editor extension was installed from the Open VSX marketplace and loaded — proven live, not just claimed.
- **Secure, fast front door.** HTTPS with a modern fast protocol (HTTP/3) and response compression; the plain-HTTP port redirects to HTTPS.

What is still in progress / needs the operator:

- **Login-page copy/paste buttons** and a **click-through marketplace test** — being built now, **not finished**.
- **Watching yourself edit in the browser** (typing, clicking files) is confirmed **by a person** for now — the automated click-through is still being wired.
- **Real public HTTPS certificate** needs a public domain and is **enabled by the operator**; the machinery is ready.
- **A host firewall tweak** (a one-line rule) needs the operator to apply it as root — the project itself never uses root.
- **Feature videos.** We do not yet have recorded "watch it work" videos; every claim above is backed by a saved test-evidence file instead. Recording the videos is a tracked to-do.

Bottom line: the new real-account SSH-key login and the editor experience are **working and proven** on the live system (23 of 23 test suites passed). A few polish items (copy/paste buttons, click-through UI test, feature videos) are still in progress and honestly marked as such.

Page 2 — For software engineers

Release: codeserver-1.0.0-dev-0.0.3. **Release-gate aggregate:** 23/23 suites PASS, 0 FAIL — qa-results/run_all/20260701T151300Z-3182121/summary.txt. **Go gate:** 70 tests, -race, 81.8% coverage.

Architecture: Browser → Caddy (containerized rootless TLS edge, HTTP/3+Brotli, forward_auth) → helix-auth (Gin gate, host-native systemd --user, loopback :8081, non-root) + code-server (host-native as milosvasic, --auth none, loopback :8080).

Live-validated (PASS + captured evidence):

- **SSH-key challenge-response auth** — services/auth_gate/{challenge,verifier,cookie,server}.go; e2e_auth 5/5, challenges_auth CH1, helixqa_auth QA-002/004; Go 70 tests -race.
- **Login redirect** — server.go handleAuth (303→/login for HTML GET, 401 otherwise), commit 75e2d9b; TestAuthBrowserNavigationRedirectsToLogin + e2e_auth (B).

- **Fail-closed** — stress_chaos_auth 5/0/1 (live gate + code-server kill+recover), challenges_auth CH2, e2e_auth (E), helixqa_auth QA-003.
- **Cookie hardening** (__Host-+HMAC+HttpOnly+Secure+SameSite=Strict, regen on login) — cookie.go; security_auth 5/5.
- **DoS hardening** (rate-limit rightmost-XFF + CSRF double-submit + spawn ceiling) — ratelimit.go, server.go; load_auth 3/0/1, concurrency_auth 5/5, race 2/2.
- **CVE-2026-35414** server-pinned principal + key-type allow-list + OpenSSH floor — verifier.go, sshversion.go, config.go.
- **Host-native editor** — deploy/systemd/helix-code-server.service; editor loads (9 markers) authed 200 + X-Helix-User. Tarball install scripts/install-auth.sh (~/.local/lib, npm fallback, pinned 4.117.0).
- **Real-user env** — ssh-key git from terminal (CH3, git ls-remote 5 refs), .bashrc exports (CH4), Explorer defaults to \$PROJECTS_ROOT (CH5, convenience, not a jail).
- **Extension install+use** — tests/types/extensions_auth.sh + bank tests/banks/helixcode-extensions.yaml; 5/5 checks captured qa-results/tests/extensions_auth/20260701T162817Z-4080597 (Open VSX 200 → load → cleanup).
- **Caddy edge** — deploy/Caddyfile, deploy/caddy/Dockerfile (xcaddy-brotli); HTTP/3+Brotli, forward_auth, /proxy 403, X-Helix-User strip; security_auth 5/5, benchmark_auth 4/4.

Non-PASS states (honest §11.4.45):

- **PENDING** — login-form clipboard buttons (assets/login_enhance.js, source landed, unvalidated); UI-driven marketplace test; edge reboot-persistence Quadlet (deploy/quadlet/helixcode-caddy.container + scripts/install-edge-boot.sh landed, arm/full-reboot survival not captured); **all Video-confirmation** (no real-use video corpus yet).
- **OPERATOR-ATTENDED** — in-browser editing (HCA-QA-UI-001 SKIP operator_attended).
- **OPERATOR-BLOCKED** — loopback firewall apply (harden-loopback.sh --apply, root); real public Let's Encrypt (public domain / DNS-01).
- **SKIP** — legacy container+password suites (integration, e2e, security, tls_letsencrypt, full_automation, concurrency, memory, benchmark, helixqa, challenges) → topology_unsupported, superseded by <name>_auth (no false-FAIL, §11.4.1).

Notes: code-server --auth none on loopback:8080 is a documented residual (auth enforced at gate/edge; harden-loopback.sh mitigates). DOCX exported via pandoc this release; no standing DOCX pipeline in tests/run_all_types.sh yet (tracked §11.4.60/§11.4.65 follow-up).